Status	Finished
Started	Saturday, 10 October 2026, 10:50 AM
Completed	Saturday, 10 October 2026, 10:54 AM
Duration	3 mins 56 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

a. Being the most efficient sorting algorithm
b. Using the divide-and-conquer approach
c. Bubbling up the largest element to its correct position with each pass
d. Sorting data in a non-sequential manner

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **2**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

a. Sorting the entire list in one pass
b. Bubbling up the largest element to its correct position
c. Dividing the list into halves
d. Bubbling up the smallest element

The correct answer is: Bubbling up the largest element to its correct position

Question **3**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. Selection
b. Searching
c. Duplicates
d. All of the above

The correct answer is: All of the above

Question **4**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Linear Search
b. Binary Search
c. Bubble Sort
d. Merge Sort

The correct answer is: Merge Sort

Question **5**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

a. Bubble Sort
b. Merge Sort
c. Insertion Sort
d. Quick Sort

The correct answer is: Bubble Sort

Question **6**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort is difficult to implement
b. Bubble Sort cannot handle duplicate elements
c. Bubble Sort does not guarantee sorted order
d. Bubble Sort is less efficient for large lists

The correct answer is: Bubble Sort is less efficient for large lists

Question **7**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Rearranging data without sorting
b. Dividing the input into parts, solving each part, and combining the solutions
c. Sorting data sequentially
d. Sorting data in a single pass

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **8**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It allows for efficient parallel processing and sorting of data
b. It simplifies the sorting process by using only one pass
c. It only works on small datasets
d. It avoids the need for recursion

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **9**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Merge Sort
b. Quick Sort
c. Bubble Sort
d. Linear Search

The correct answer is: Merge Sort

Question **10**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 80,150,200,470,40,90,300,400,70,120
- b. 40,80,90,150,200,300,400,470,70,120
- c. 200,470,80,150,40,90,300,400,70,120
- d. 40,70,80,90,120,150,200,300,400,470

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **11**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

- a. Each part is searched for a specific element
- b. The parts are merged without sorting
- c. The parts are ignored
- d. Each part is sorted independently

The correct answer is: Each part is sorted independently

Question **12**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

- a. A sorting algorithm is stable if it preserves the order of all keys
- b. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys
- c. A sorting algorithm is stable if it preserves the order of non-duplicate keys
- d. A sorting algorithm is stable if it preserves the order of duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **13**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Arranging data in a specific order
- b. Finding a specific element in a list
- c. Merging two datasets
- d. Removing duplicates from a list

The correct answer is: Arranging data in a specific order

Question **14**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

- a. Linear Search
- b. Binary Search
- c. Merge Sort
- d. Bubble Sort

The correct answer is: Merge Sort

Question **15**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. order()
- b. sorted()
- c. arrange()
- d. sort()

The correct answer is: sorted()